ANALYSIS OF PLACEMENT DATASET

Measure Of Location Of Data:

1. Percentile

```
descriptive = pd.DataFrame(index=["Mean", "Median", "Mode", "Q1:25%", "Q2:50%", "Q3:75%", "99%", "Q4:100%"], columns=Quan)
for ColumnName in Quan:
    descriptive[ColumnName]["Mean"]=dataset[ColumnName].mean()
    descriptive[ColumnName]["Median"]=dataset[ColumnName].median()
    descriptive[ColumnName]["Mode"]=dataset[ColumnName].mode()[0]
    descriptive[ColumnName]["Q1:25%"]=dataset.describe()[ColumnName]["25%"]
    descriptive[ColumnName]["Q2:50%"]=dataset.describe()[ColumnName]["50%"]
    descriptive[ColumnName]["Q3:75%"]=dataset.describe()[ColumnName]["75%"]
    descriptive[ColumnName]["99%"]=float(np.percentile(dataset[ColumnName],[99]))
    descriptive[ColumnName]["Q4:100%"]=dataset.describe()[ColumnName]["max"]
```

```
descriptive
```

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Mean	108.0	67.303395	66.333163	66.370186	72.100558	62.278186	288655.405405
Median	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Mode	1	62.0	63.0	65.0	60.0	56.7	300000.0
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Q4:100%	215.0	89.4	97.7	91.0	98.0	77.89	940000.0

SSC Percentage:

- Q1 = **60.6** meaning 25% of the students scored below these values, which identifies the lower-range performance level of the students.
- Q2 = **67** (Median) indicating typical performance is around this level and the increasing percentage between Q1 and Q2 is 7%.
- Q3 = **75.7**, the 75th percentile highlights higher-performing students
- Maximum score is **89.4**, suggesting some high performers.

HSC Percentage:

The majority of the students scored between **60.9** to **73.** This indicates moderate variation between **91.86** and **97.7** with some very high achievers.

DEGREE Percentage:

Shows a fairly consistent distribution with gradual improvement at higher percentiles. Most students scored between the range **61 to 72.** The **99th percentile** reflects extreme high-end values, where the student scored 83.86. The students performing **100th percentile** represents the maximum observed value.

ENTRANCE TEST Percentage:

The Entrance test suggests wider dispersion compared to earlier academic scores. Most of the students scored 60% to 83.5% (Q1–Q3 range), the second quartile score was 71%, while top performers scored between 97% to 98%.

MBA Percentage:

The first quartile (25th percentile) is 57.945%, meaning one quarter of the students obtained marks lower than this level. The second quartile score is 62%, which shows that half of the students scored below this value and the other half scored above it, while the third quartile (75th percentile) is 66.255%, indicating that three quarters of students scored below this mark. The maximum observed score is 77.89%.Overall, the percentile spread demonstrates that MBA scores are clustered fairly closely around the central values.

Salary Distribution:

The Q1 salary is ₹240,000 and Q3 salary is ₹300,000, this indicates that half of the individuals earn within this range. The Q2 median salary is ₹265,000, which represents the typical compensation level. The Maximum salary is ₹940,000. Significantly higher than Q3, suggesting presence of high-end outliers or top performers. The mean salary (₹288,655) being higher than median indicates a right-skewed distribution, influenced by high salaries.

2. IQR-Inter Quartile Range:

The Interquartile Range (IQR) is the difference between the third quartile(Q3) and first quartile(Q1), representing the spread of the middle 50% of a dataset.

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Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q2:50%	108.0	67.0	65.0	66.0	71.0	62.0	265000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
99%	212.86	87.0	91.86	83.86	97.0	76.1142	NaN
Q4:100%	215.0	89.4	97.7	91.0	98.0	77.89	940000.0
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5rule	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
Lower_Bound	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
Upper_Bound	322.0	98.35	91.15	88.5	118.75	78.72	390000.0
Min	1	40.89	37.0	50.0	50.0	51.21	200000.0
Max	215	89.4	97.7	91.0	98.0	77.89	940000.0

SSC_P Marks:

The lower bound (37.95) and upper bound (98.35) falls within the min and max range for the SSC marks. Hence no outliers in the SSC_P marks.

HSC_P:

The lower bound is 42.75 and the upper bound is 91.5. Both the values exceed the min 37.0 and max 97.7 scores. This indicates the presence of outliers in HSC_P marks.

DEGREE_P:

The lower bound value is 44.5 and it is within the min value 50.0. The max score 91.0 exceeds the upper bound value 88.5

Outliers present in the Degree_P marks.

Etest_P:

The lower bound (24.75) is below the min value (50.0). The upper bound value (118.75) is above the max value (98.0).

This indicates there are no outliers in Etest_P scores.

MBA_P:

The lower bound is 45.48 and the upper bound is 78.72. Both the values fall within the min 51.21 and max 77.89 scores.

This proves that there are no outliers in MBA_P marks.

Salary:

The lower bound 150000 is below the min value 200000. The max salary value 940000 exceeds the upper bound salary 390000.

This indicates the presence of outliers in the Salary column.